# Supplementary Material for: Federated O-RAN Slice SLA Prediction: A Cross-Architecture Empirical Benchmark on Colosseum/ColO-RAN

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Cross-architecture federated-learning benchmark on Colosseum/ColO-RAN: per-section appendix material moved out of the main paper for length compliance with IEEE journal main-body conventions. Each section heading carries the originating main-paper section in parentheses.

## APPENDIX A
## MECHANISM ANALYSIS DETAILS (EXTENDS § 7.1)

### A. Applicability boundary (extends § 7.1.3)

The empirical finding (natural-by-BS dominance and inverted $\alpha_{\text{dir}}$ monotonicity) is dataset-structural, not algorithm-mechanistic. We therefore expect it to replicate whenever a FL deployment satisfies (a) clients are partitioned along an axis on which the dataset has measurable distributional heterogeneity (here: `bs_id`), (b) the alternative partition strategy redistributes rows along an orthogonal axis that *does not* correspond to dataset structure (here: `slice_id`, which is statistically uniform across `bs_id`s in this corpus). Conversely, datasets where the natural client partition has uniform marginals on every axis would be expected to behave classically (lower $\alpha_{\text{dir}} \Rightarrow$ harder). We conjecture similar inversion will appear in other near-RT RIC forecasting workloads (uplink throughput-violation prediction, latency-budget breach prediction) where per-gNB telemetry distributions differ measurably; quantitative verification on additional corpora is open future work.

### B. Hardware drift caveat for the § 7.1.1 ablation (extends § 7.1.4)

The § 7.1.1 `random_split` ablation cells were run on $4\times$ Tesla V100-SXM2-32GB (sm_70, driver 535.161, CUDA 12.1) while the Phase 5 baseline (natural-by-BS and Dirichlet partitions) was run on RTX 4080 (sm_89). The two GPU generations have different bf16 implementations (V100 emulates bf16 via fp16 tensor cores with possible fp32 fallback; Lovelace has native bf16 tensor cores), and `cudnn_deterministic=True` pins kernel choice within a hardware target but does not impose bit-equivalence across hardware. We bound the resulting AUC drift by comparing V100 `random_split` AUC to 4080 Dirichlet $\alpha_{\text{dir}} = 5.00$ AUC (both are partition strategies that destroy bs grouping, so they should yield equivalent AUC under hardware-independent training): the residual delta is at most $0.0072$ (LSTM), $0.0043$ (Mamba), $0.0021$ (Spiking-SSM), all within the corresponding seed $\sigma$. Hardware drift is therefore bounded above by $\sim 0.007$ AUC, more than an order of magnitude smaller than the $\sim 0.18$ mechanism signal (ratio $\approx 25\times$). Strict elimination of hardware confound would require re-running Phase 5's IID column on V100 ($\approx 10$ GPU-hours on the $4\times$ cluster); we did not undertake this because the bounded-drift argument is sufficient to interpret § 7.1.1's ablation conclusion.

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## APPENDIX B
## REPRODUCIBILITY INFRASTRUCTURE DETAIL (EXTENDS § 5)

### A. Test infrastructure (extends § 5.2)

The release ships **277 passing tests** across four suites that protect numerical and pipeline-level invariants:

- `tests/test_v7_*.py` — 202 unit + integration tests for `fl_v7` trainer (model-build, partition correctness, scaler determinism, FedDyn canonical-vs-Option-II contract).
- `tests/test_aggregate_v7_results.py` — 22 tests for the aggregator (paired-bootstrap, BLAKE2b decorrelation, JSON round-trip, schema-mismatch detection, 1260-cell scaling).
- `tests/test_paper_draft_invariants.py` — 37 paper-text invariants (forbidden hallucinated facts; required corrected facts; license consistency; hardware specs).
- `tests/test_paper_claims_sources.py` — 16 paper-vs-data claims tests that re-read `artifacts/step1_factfinding.json`, `step2_mechanism_search.json`, and `aggregated_phase5.json` and assert paper-reported numbers match the underlying measurements within rounding tolerance.

The latter two suites (53 tests total) form a paper-correctness developer pre-commit gate: every `PAPER_DRAFT.md` edit is verified by re-running them, preventing quick-claim regressions.

### B. Statistical pipeline (extends § 5.3)

`scripts/aggregate_v7_results.py` consumes the per-cell `summary.json` files and produces `aggregated_phase5.json` (90 group means + 270 paired-bootstrap distributions: 180 algorithm-pairs + 90 architecture-pairs) plus a paper-grade Markdown table. Each pairwise comparison's bootstrap RNG seed is derived from a BLAKE2b hash of the pair identifier added to a base seed (2026 for algorithm-pairs, 2027 for arch-pairs), guaranteeing independent bootstrap streams across all 270 distributions and supporting joint coverage claims (§ 4.5).

### C. Croissant dataset metadata (extends § 5.4)

The Colosseum/ColO-RAN public release is upstream-licensed; our preprocessed `coloran_raw_unified.parquet` ($\approx$ 18 M rows) and the partition specifications are documented in a Croissant 1.0 metadata file (`croissant.json` at the repository root, validated against the official `mlcroissant` validator) shipped with the release archive. The metadata block describes the (`bs_id`, `slice_id`, `sched`, `tr`) keying, the 17 continuous features (§ 3.1), the OOD train/val/test split (§ 3.3), and the `add_classification_target` derivation rule (§ 3.2).

### D. Reproducible execution environment (extends § 5.5)

The release archive bundles:

- `Dockerfile.repro` — pinned to the exact CUDA / PyTorch / `nvidia-ml-py` versions used in Phase 5 (CUDA 12.8, PyTorch 2.10, `nvidia-ml-py` 12.x);
- `requirements.lock` — SHA256-pinned Python dependencies (74 packages);
- `repro.sh` — one-line entry point that loads the parquet, runs a 1-cell smoke (LSTM, FedAvg, IID, `seed=42`, 5 rounds $\times$ 20 max-steps, $\sim$ 30 s on RTX 4080), and verifies bit-equivalent test AUC against a stored reference value within $10^{-6}$.

### E. Demo notebook (extends § 5.6)

A Jupyter notebook (`notebooks/colosseum_oran_federated_slicing_demo.ipynb`) walks through (a) loading aggregator JSON, (b) reproducing Figures 1–3, (c) running a downscaled smoke version of the § 7.1.1 `random_split` ablation on a single GPU (the full ablation runs on the 4$\times$ V100 setup documented in App. A-B), and (d) regenerating § 6 paired-bootstrap $\text{CI}_{95}$ from raw cells. The notebook is the recommended onboarding entry point for downstream researchers extending the benchmark.

## APPENDIX C
## EXTENDED LIMITATIONS AND THREATS TO VALIDITY (EXTENDS § 8)

Each subsection below holds the full prose of an § 8 limitation whose main-paper bullet was condensed to a 1–2 sentence headline. The § 1 contribution(s) each item threatens are stated in the main-paper § 8 bullet headers.

**A note on internal references in C.1–C.6:** "Stage 1" refers to the predecessor centralized 3-architecture benchmark on ColO-RAN (LSTM, Mamba, Spiking-SSM) whose pre-registered hardware budget, precision policy, and gradient-step count are inherited by Phase 5 (the federated extension reported in this paper). "ADR-001 D-N" denotes the $N$-th preregistered design decision in the project's Architecture Decision Record. Both artefacts are documented in the release archive for downstream reproduction.

### A. L9 full discussion: `pos_weight_split=train` choice (extends § 8 L9)

We compute the BCE positive-class weight from the train partition's positive rate (per ADR-001 D-12), in line with standard practice that prevents test-set positive-rate from leaking into the loss. Two alternative computations (val-derived `pos_weight`; held-out-fold `pos_weight`) would also be valid and might shift absolute AUC numbers slightly, but per § 7.1 our V100 `random_split` ablation confirms the inverted-$\alpha_{\text{dir}}$ mechanism direction is robust to the partition-axis shuffle, and the BCE loss reweighting is identical (globally pooled) across all clients regardless of which split the rate is computed from — so the *direction* of all our findings is invariant to this choice.

### B. L10 full discussion: bf16 mixed-precision training versus fp32 (extends § 8 L10)

All Phase 5 cells use bf16 mixed precision per the Stage 1 inheritance contract for memory and throughput on RTX 4080. The numerical-precision gap between bf16 and fp32 is bounded for our model architectures by the lack of overflow-prone operations (no exponential losses, no large-fan-in matmul without LayerNorm); we do not report a fp32 verification cell and treat bf16 vs fp32 cross-check as a reproducibility-supplementary item for the camera-ready archive. Readers reproducing on hardware without native bf16 (Pascal sm_60 and earlier) should expect $O(0.001)$ AUC drift.

### C. L11 full discussion: per-client compute budget split (extends § 8 L11)

Our (round, max_steps) configuration matches Stage 1's audit-corrected 25 000 total-gradient-steps budget. An alternative split (e.g., 50 rounds $\times$ 100 max-steps, 200 rounds $\times$ 25 max-steps) would shift the round-vs-step ratio and potentially affect server-side aggregation count by 2–4$\times$; we did not sweep this dimension, treating round count as a fixed reproducibility anchor. Whether the FedAdam-saturates-headroom finding (§ 6.3) survives at 4$\times$ more rounds with proportionally fewer steps per round is open.

### D. L12 full discussion: SLA-threshold sensitivity (extends § 8 L12)

The 10% BLER threshold is the canonical ColO-RAN SLA gate [1]; the empirical positive rate at this threshold on the train partition is 30.9% (§ 7.1 setup-level facts). A threshold sweep over 5% / 15% / 20% would test whether the inverted-$\alpha_{\text{dir}}$ monotonicity strengthens at sparser-positive regimes (where any structural specialisation has higher loss-leverage) and weakens at higher thresholds; we did not run this sweep but predict (a) inversion strengthens as threshold tightens and (b) inversion weakens or disappears as threshold approaches the symmetric $50\%$ regime. This is the highest-leverage open ablation for the camera-ready revision after § 7.1.1.

### *E. L13 full discussion: bootstrap CI percentile vs BCa (extends § 8 L13)*

§ 4.5 explains the choice: at our $n = 10$ paired seeds with approximately symmetric per-seed delta distributions, the bias-correction term in the BCa interval is bounded above by $O(1/\sqrt{n}) \approx 0.32$ standard errors; this magnitude is smaller than our seed $\sigma$ on every reported finding except § 6.4's Mamba×SCAFFOLD interaction, where $\sigma$ inflation by $23\times$ makes the CI choice less consequential than the Wilcoxon $p$-value. Readers who prefer BCa should treat our $\text{CI}_{95}$ as a slightly tighter version of the BCa interval; the qualitative conclusions are unchanged.

### *F. L14 full discussion: FedAdam hyperparameter sensitivity (extends § 8 L14)*

$\beta_1 = 0.9$, $\beta_2 = 0.99$, $\eta_{\text{server}} = 0.01$ follow the FedAdam paper's preregistered values [2]; $\beta_2 = 0.99$ differs from canonical Adam's $0.999$. We did not run a $(\beta_1, \beta_2, \eta_{\text{server}})$ sensitivity sweep — the FedAdam-vs-FedAvg deltas reported in § 6.3 are anchored to paper-canonical hyperparameter choices rather than to a per-task hyperparameter optimum. Whether the FedAdam-saturates-headroom finding survives at alternative $\beta_2$ (e.g., $0.999$) is open; we expect modest sensitivity ($\leq 0.005$ AUC drift) given Adam's relative robustness to $\beta_2$ variation in offline ML literature, but this is anticipated, not measured.

## APPENDIX D
## FUTURE WORK DIRECTIONS (EXTENDS § 9.1)

Several open questions remain. **First**, § 7.1.5 disambiguates the inverted-$\alpha_{\text{dir}}$ mechanism (the strong reading of § 7.1.1 (i) — bs grouping alone suffices — is refuted; (ii) per-client structural coherence is the additional necessary ingredient), but two design constraints (`sub_per_bs=2` halving per-client data, $5/14$ sampling fraction vs Phase 5's $5/7$) prevent a clean decomposition of the 1–4 pp residual gap to natural-by-BS; an isolation re-run keeping `sub_per_bs=2` while bumping `clients-per-round` to 10 (matches Phase 5's $5/7$ sampling fraction at the doubled client count) would isolate the slice-mixing effect from per-round participation, following the non-IID-impact assessment methodology of Jimenez-Gutierrez et al. [3] and the proximity-partition framing of ProFed [4]. **Second**, our $10\%$ BLER threshold should be perturbed to $\{5, 15, 20\}\%$ to confirm operational robustness, in line with recent rare-event metric-stability work [5]. **Third**, scaling beyond $N = 7$ base stations to the larger-$N$ 6G slicing and edge-FL regimes studied by Chiarani et al. [6] and Sezgin et al. [7] would test whether inverted-$\alpha_{\text{dir}}$ is a small-$N$ artefact. **Fourth**, the FedAdam headroom-saturation finding may be soft: variance-aware sharpness methods such as FedSCAM [8] and synthetic-data-guided FedSynSAM [9] introduce per-client perturbation modulation that could extract additional accuracy. **Fifth**, integrating differential privacy via over-the-air channel noise [10] and exploring neuromorphic edge deployment — both sparse-RNN acceleration on Loihi-class accelerators [11] and spiking-NN low-power edge sensing [12] — would close the loop toward production-deployable privacy-preserving energy-efficient O-RAN slice prediction.

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